

Exam 2 Curve

$$f(x) = \begin{cases} 70 - \frac{7}{13}(70-x) & x > 31 \\ x + 18 & x \leq 31 \end{cases}$$

Positive Definiteness

Theorem 20.1 If A is self-adjoint, i.e. that $(A)_{ij} = \overline{(A)_{ji}}$, then $\langle \vec{x}, A\vec{x} \rangle \in \mathbb{R}$ for all $\vec{x}$.

$$\langle \vec{x}, A\vec{x} \rangle = \langle A\vec{x}, \vec{x} \rangle = \overline{\langle \vec{x}, A\vec{x} \rangle}$$

Def 20.2 A self-adjoint matrix A is said to be positive definite iff $\langle \vec{x}, A\vec{x} \rangle > 0$ for all $\vec{x} \neq \vec{0}$.

Example: The matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix}$ is positive def.

$$\begin{aligned} \langle \vec{x}, A\vec{x} \rangle &= [x_1 \ x_2] \begin{bmatrix} 1 & 2 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \\ &= [x_1 \ x_2] \begin{bmatrix} x_1 + 2x_2 \\ 2x_1 + 6x_2 \end{bmatrix} \end{aligned}$$

$$= x_1^2 + 2x_1x_2 + 2x_1x_2 + 6x_2^2$$

$$= x_1^2 + 4x_1x_2 + 6x_2^2$$

Now geometrically $x_1^2 + 4x_1x_2 + 6x_2^2 = a$ is an ellipse for $a > 0$ and has no solutions for $a < 0$.

$$(A^* = A)$$

Theorem 20.3 Let A be self-adjoint. Then

TFAE (the following are equivalent)

- i. A is positive definite
- ii. $\langle \vec{x}, A\vec{x} \rangle > 0$ for $\vec{x} \neq \vec{0}$
- iii. The e-values of A are strictly positive
- iv. The k -minor matrix,

$$A_k = [(A)_{ij}]_{i=1, \dots, k}^{j=1, \dots, k}$$

$$\left[\begin{matrix} A_k \\ \vdots \\ A \end{matrix} \right]_{n \times n}$$

has positive determinant for each k

- v. $A = LL^*$ for some lower triangular matrix L

Def 20.4 If A is positive definite, then the factorization, $A = LL^*$ is the Cholesky factorization.

Proof 20.3: Let A be self adjoint and set

$$A = \sum \lambda_i \vec{e}_i \vec{e}_i^* . \text{ I.e. } A \vec{e}_i = \lambda_i \vec{e}_i \text{ and}$$

$$\langle \vec{e}_i, \vec{e}_j \rangle = \delta_{ij}$$

ii $\Leftrightarrow$ iii :

$\Rightarrow$ Assuming that $\langle \vec{x}, A\vec{x} \rangle > 0$ for all $\vec{x} \neq \vec{0}$ we have that

$$\begin{aligned} \lambda_i &= \lambda_i \langle \vec{e}_i, \vec{e}_i \rangle \\ &= \langle \vec{e}_i, \lambda_i \vec{e}_i \rangle \\ &= \langle \vec{e}_i, A\vec{e}_i \rangle > 0 \end{aligned}$$

$\Leftarrow$ Assuming that $\lambda_i > 0$ for each i we have that for $\vec{x} = \sum x_i \vec{e}_i$

$$\begin{aligned} \langle \vec{x}, A\vec{x} \rangle &= \left\langle \sum_i x_i \vec{e}_i, A \sum_j x_j \vec{e}_j \right\rangle \\ &= \left\langle \sum_i x_i \vec{e}_i, \sum_j x_j \lambda_j \vec{e}_j \right\rangle \end{aligned}$$

$$= \sum_i \sum_j \bar{x}_i x_j \lambda_j \langle \vec{e}_i, \vec{e}_j \rangle$$

$$\begin{cases} 1 & i=j \\ 0 & i \neq j \end{cases}$$

$$= \sum_j \bar{x}_j x_j \lambda_j$$

$$= \sum_j \|x_j\|^2 \lambda_j > 0$$

ii $\Leftrightarrow$ iv

Note that $\frac{|A_k|}{|A_{k-1}|}$ is the k^{th} pivot

of A in the LDU factorization

ii $\Leftrightarrow$ v $(AB)^* = B^* A^*$

Since $A^* = A$ we have that for

$A = LDU$, $U = L^*$. Furthermore

as A is positive definite, D is

also positive definite and $\text{diag } D$ is

strictly positive. Now setting $\sqrt{D}$

to be such that

$$\text{diag } \sqrt{D} = \{+\sqrt{D_{1,1}}, \dots, +\sqrt{D_{n,n}}\}$$

we have that

$$\begin{aligned} A &= L \sqrt{D} \sqrt{D} L^* \\ &= (L \sqrt{D})(L \sqrt{D})^* \end{aligned}$$

Exercise :

1) show that $A = \begin{matrix} & \overset{A_1}{\begin{bmatrix} 2 & 2 \\ 2 & 5 \end{bmatrix}} & \overset{A_2}{\begin{bmatrix} 0 \\ 3 \end{bmatrix}} & \overset{A_3}{\begin{bmatrix} 0 \\ 3 \\ 8 \end{bmatrix}} \end{matrix}$ is

positive definite and find the Cholesky factorization.

Since the k -minors of A have determinants

$$|A_1| = 2$$

$$|A_2| = \begin{vmatrix} 2 & 2 \\ 2 & 5 \end{vmatrix} = 6$$

$$|A_3| = \begin{vmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{vmatrix} = 30$$

which are all positive, A is pos. def.

Now finding the Cholesky :

$$\begin{bmatrix} 2 & 2 & 0 \\ 2 & 5 & 3 \\ 0 & 3 & 8 \end{bmatrix} \xrightarrow{-r_1} \begin{bmatrix} 2 & 2 & 0 \\ 0 & 3 & 3 \\ 0 & 3 & 8 \end{bmatrix} \xrightarrow{-r_2}$$

$$\Rightarrow \begin{bmatrix} 2 & 2 & 0 \\ 0 & 3 & 3 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & & \\ & 3 & \\ & & 5 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\therefore A = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 2 & & \\ & 3 & \\ & & 5 \end{bmatrix}}_D \underbrace{\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}}_{U=L^T}$$

$$= \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{3} & \\ & & \sqrt{5} \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{3} & \\ & & \sqrt{5} \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Thus $A = LL^*$ where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} \sqrt{2} & & \\ & \sqrt{3} & \\ & & \sqrt{5} \end{bmatrix} \\ = \begin{bmatrix} \sqrt{2} & 0 & 0 \\ \sqrt{2} & \sqrt{3} & 0 \\ 0 & \sqrt{3} & \sqrt{5} \end{bmatrix}$$

Def 20.5 Let A be self-adjoint. Then

A is ,

i. Pos. Def. if $\langle \vec{x}, A\vec{x} \rangle > 0$ for $\vec{x} \neq \vec{0}$

- ii. Pos. semi-def. if $\langle \vec{x}, A\vec{x} \rangle \geq 0$ for $\vec{x} \neq \vec{0}$
- iii. Negative definite if $\langle \vec{x}, A\vec{x} \rangle < 0$ for $\vec{x} \neq \vec{0}$
- iv. Negative semi-definite if $\langle \vec{x}, A\vec{x} \rangle \leq 0$ for $\vec{x} \neq \vec{0}$
- v. Indefinite if $\langle \vec{x}, A\vec{x} \rangle$ can be both positive and negative

Theorem 20.6 let A be self-adjoint. Then

TFAE

- i. A is negative definite
- ii. $\langle \vec{x}, A\vec{x} \rangle < 0$ for $\vec{x} \neq \vec{0}$
- iii. The e-values of A are all negative
- iv. The k -minor matrices satisfy $|A_1| < 0$ and $|A_{k+1}|/|A_k| < 0$ for $k \geq 1$
- v. The pivots in the LDU factorization are all negative

$$A \rightarrow \begin{bmatrix} -d_1 & & * \\ & \ddots & \\ 0 & & -d_n \end{bmatrix}$$

Example : $A = \begin{bmatrix} -3 & 2 \\ 2 & -4 \end{bmatrix}$ is negative definite

$$\text{as } |A_{11}| = -3 \text{ and } \frac{|A_{22}|}{|A_{11}|} = \frac{\begin{vmatrix} -3 & 2 \\ 2 & -4 \end{vmatrix}}{-3} = \frac{8}{-3}$$

$$\text{or as } \lambda = \frac{-7 \pm \sqrt{17}}{2} < 0.$$

(Also $\begin{bmatrix} -3 & 2 \\ 2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} (-3) & 2 \\ 0 & (-8/3) \end{bmatrix}$, pivots both negative)

Theorem 20.7 (2^{nd} Derivative test for multivariable functions)

* $\vec{x}_0$ is a critical point of f

let $f = f(\vec{x})$ be analytic in the ball

$\{\|\vec{x} - \vec{x}_0\| < r\} \subseteq \mathbb{R}^n$ and let $\nabla f(\vec{x}_0) = \vec{0}$. *

Then for any $\vec{y}$ s.t. $\|\vec{y} - \vec{x}_0\| < \varepsilon$ where $\varepsilon < r$ we have that -

$$f(\vec{y}) = f(\vec{x}_0) + \underbrace{\frac{1}{2} \langle \vec{y} - \vec{x}_0, A(\vec{y} - \vec{x}_0) \rangle}_{\text{quadratic part}} + \underbrace{O(\varepsilon^3)}_{\text{remainder}}$$

where $O(\varepsilon^3) \rightarrow 0$ as $\varepsilon \rightarrow 0$ and A

is the Hessian matrix $A_{i,j} = \frac{\partial^2 f(\vec{x}_0)}{\partial x_i \partial x_j}$

Thus,

See
Figure 1
below

- i. $f(\vec{x}_0)$ is a local max if A is negative definite
- ii. $f(\vec{x}_0)$ is a local min if A is positive definite
- iii. $f(\vec{x}_0)$ is a saddle if A is indefinite

Exercise:

2) The quadratic

$$f(\vec{x}) = 2x_1x_2 + 2x_2x_3 - 2x_1^2 - 3x_2^2 - 4x_3^2$$

has a critical point at the origin.

Determine whether it is a local max, local min or saddle.

$$\text{Since } f_{x_1x_1} = -4, f_{x_1x_2} = 2, f_{x_1x_3} = 0$$

$$f_{x_2x_2} = -6, f_{x_2x_3} = 2, f_{x_3x_3} = -8$$

we have that $f = \frac{1}{2} \langle \vec{x}, A\vec{x} \rangle$ where

$$A = \begin{bmatrix} -4 & 2 & 0 \\ 2 & -6 & 2 \\ 0 & 2 & -8 \end{bmatrix}$$

Now as the determinants of the minors satisfy $|A_1| = -4$, $|A_2| = \begin{vmatrix} -4 & 2 \\ 2 & -6 \end{vmatrix} = 20$,

$$|A_3| = |A| = \begin{vmatrix} -4 & 2 & 0 \\ 2 & -6 & 2 \\ 0 & 2 & -8 \end{vmatrix} = -4(44) - 2(-16) \\ = -144$$

such that $\frac{|A_2|}{|A_1|} = -5$ and $\frac{|A_3|}{|A_2|} = -7.2$

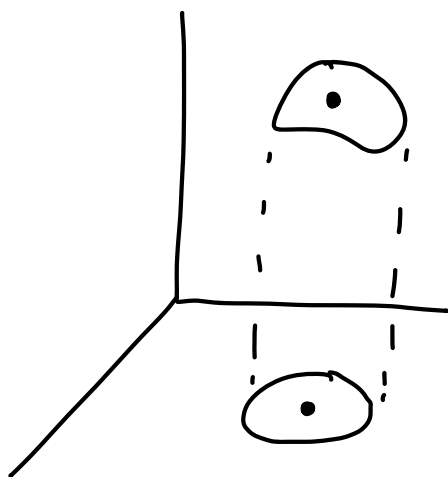
we have A is negative definite and thus the origin is a local max.

Also note that the e -values satisfy

$$\lambda = -6, -6 \pm 2\sqrt{3}$$

which are all negative

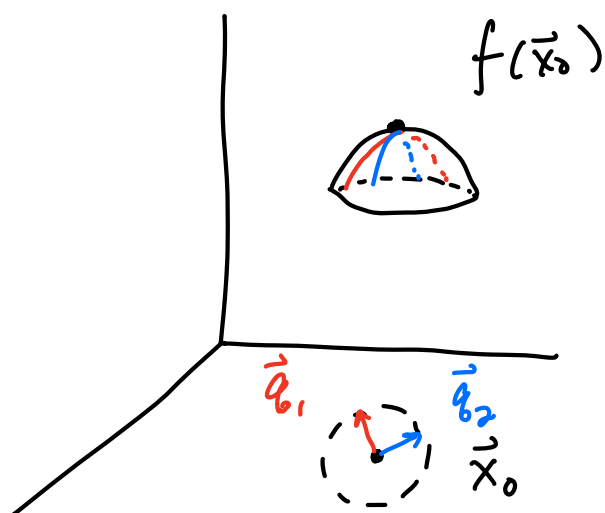
Figure 1: Critical point on a smooth surface



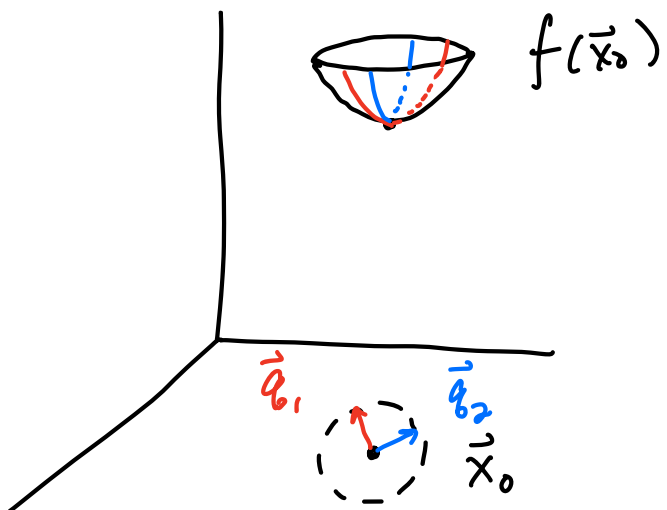
Let $\vec{x}_0$ be a critical point of f and let

$$A = \begin{bmatrix} f_{x_1 x_1}(\vec{x}_0) & f_{x_1 x_2}(\vec{x}_0) \\ f_{x_1 x_2}(\vec{x}_0) & f_{x_2 x_2}(\vec{x}_0) \end{bmatrix} \text{ be the Hessian}$$

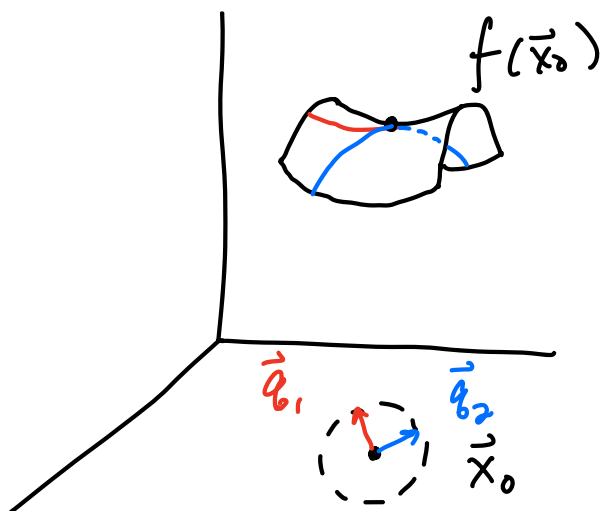
such that $A = \lambda_1 \vec{e}_1 \vec{e}_1^T + \lambda_2 \vec{e}_2 \vec{e}_2^T$



$$\lambda_1, \lambda_2 < 0$$



$$\lambda_1, \lambda_2 > 0$$



$$\lambda_1 > 0$$

$$\lambda_2 < 0$$